Matthew Cassini

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Education

New Jersey Institute of Technology (NJIT) Expected May 2028
Ph.D. in Applied Mathematics, GPA 3.938
Dissertation Advisor: Thi-Phong Nguyen

New Jersey Institute of Technology (NJIT) May 2023
B.S. in Computational Mathematics

Experience

- **Graduate Teaching Assistant**, Recitation Leader, Department of Mathematical Sciences, NJIT, Fall 2025 – Present
  - Lead recitations for Calculus I, Calculus II and Calculus II Honors
- **Graduate Research Assistant**, Prof. Thi-Phong Nguyen, Department of Mathematical Sciences, NJIT, Spring 2024 – Present
  - Created a numerical solver for the direct scattering problem of Maxwells equations on a biperiodic infinite layer with a local defect contained in one of the periods through the use of Floquet-Bloch transform and a volume integral equation
  - Studying inverse scattering problem
- **Graduate Teaching Assistant**, NJIT, Spring 2024
  - Tutored students on various mathematical subjects and MATLAB programming
- **Undergraduate Research Assistant**, NJIT, May 2022 - May 2023
  - Created a numerical solver for a non-local Monge-Ampére equation utilizing a novel discretisation technique involving the Dirac Delta distribution and monotone finite differences

Publications

- Cassini, M.A., Hamfeldt, B.F. Numerical Optimal Transport from 1D to 2D Using a Non-local Monge-Ampère Equation. La Matematica 3, 509–535 (2024).

Conference and Seminar Presentations

- (Poster) Volume Integral Method for Electromagnetic Equations, Dana Knox Research Showcase, NJIT, April 2025
- (Poster) Volume Integral Method for Electromagnetic Equations, SIAM New York-New Jersey-Pennsylvania Section Conferences (SIAM-NNP), Rochester Institute of Technology (RIT), USA, November 2024
- (Talk) Volume Integral Method for Electromagnetic Equations, HPC Research Symposium, NJIT, July 2024
- (Poster) Optimal Transport Between Unequal Dimensions, Frontiers of Applied and Computational Mathematics (FACM), NJIT, May 2023

Skills

Programming Languages: Highly proficient in MATLAB and Python. Knowledgable in C++, Mathematica and SQL.

Other Softwares: Linux Systems, Git, Slurm Workload Manager, LaTeX

Languages: Proficient in Spanish